

Auteur: Xander De Blauwer

Studierichting: Informatica

Oefeningenreeks 6F nr. 4

$$f_n : [1, +\infty] \rightarrow \mathbb{R} : x \rightarrow \frac{1}{nx^2}$$

$$\forall x \in \mathbb{R} : \lim_{n \rightarrow \infty} \left(\frac{1}{nx^2} \right) = 0 \Rightarrow \text{puntgewijs convergent}$$

$$\sup_{x \in [1, +\infty[} \left| \frac{1}{nx^2} - 0 \right| = \sup_{x \in [1, +\infty[} \left| \frac{1}{nx^2} \right| = \frac{1}{n}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \|f_n - f\|_{\infty} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \Rightarrow \text{uniform convergent}$$

References